Department of Computer Science and Engineering
BRAC University
CSE 260: Digital Logic Design

Experiment # 1: Familiarization with *Fundamental Logic Gates*

Objective:

- To get familiarized with fundamental logic gates and demonstrate the input-output relationship of 2-input **AND** (IC – 7408), **OR** (IC – 7432) **and NOT/Inverter** (IC – 7404) gates by constructing their truth tables.
- To get familiar with other logic gates like **NAND** (IC – 7400), **NOR** (IC – 7402), **XOR** (IC – 7486) and **XNOR** (IC – 4077)

Required Components:

1. IC 7408 × 1
2. IC 7432 x 1
3. IC 7404 × 1
4. IC 7400 x 1
5. IC 7402 × 1
6. IC 7486 x 1
7. IC 4077 x 1

Procedure:

- For each of the ICs, place the IC correctly on the trainer board
- Remember to connect each IC's VCC pin to the "+5V" position of the DC Power Supply of the trainer board, and the GND or 0V pin to the "GND" position of the trainer board.
- Connect the inputs to the data switches and the output to any position on the LED display.
- Find out the outputs for all possible combinations of input states.
- Write down the input-output in tabular form.

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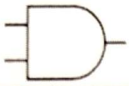
Logic gate symbols with corresponding truth tables:

NOT

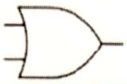
INPUT	OUTPUT
A	
0	1
1	0

AND



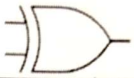
INPUT		OUTPUT
A	B	
0	0	0
1	0	0
0	1	0
1	1	1

OR



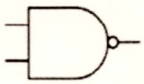
INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	1

XOR



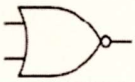
INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	0

NAND



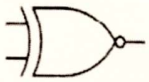
INPUT		OUTPUT
A	B	
0	0	1
1	0	1
0	1	1
1	1	0

NOR



INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	0

XNOR



INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	1

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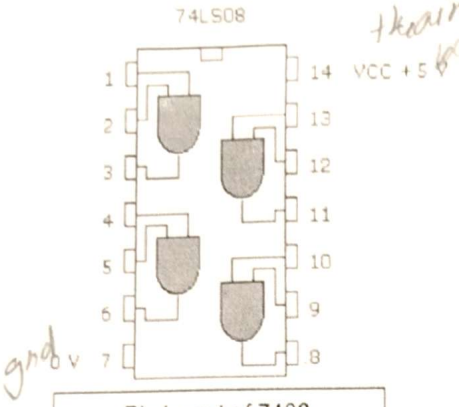
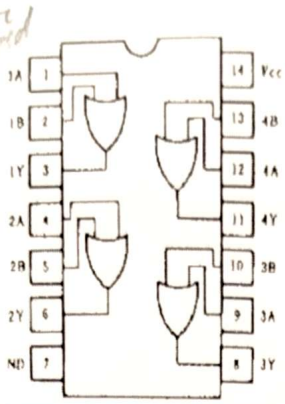
Pin Diagrams of ICs:



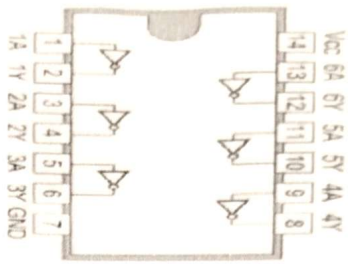
Pin layout of 7408

AND

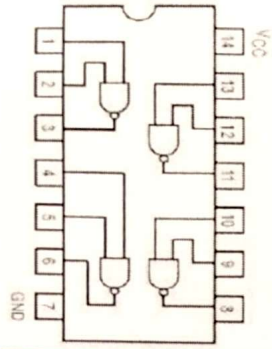


Pin layout of 7432

or

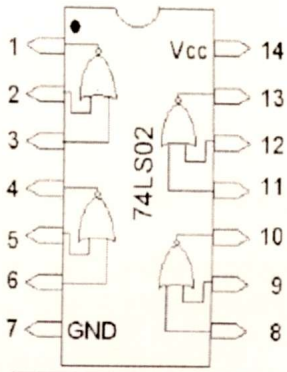


Pin layout of 7404



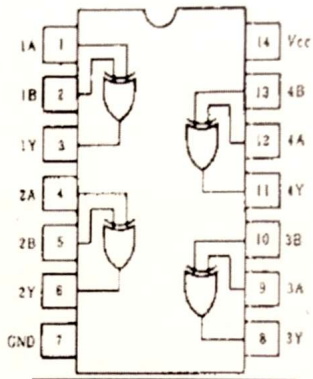
Pin layout of 7400

NAND



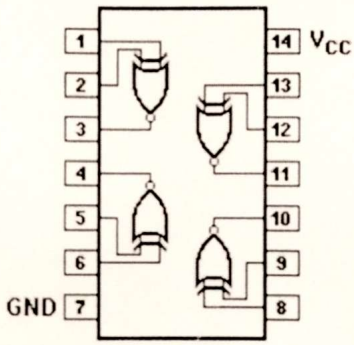
Pin Layout of 7402

NOR



Pin layout of 7486

XOR



Pin layout of 4077

XNOR

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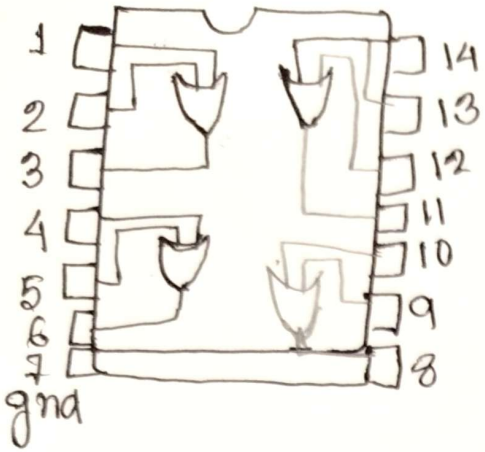
Mohammed Sabbir Ahmed - 24141020

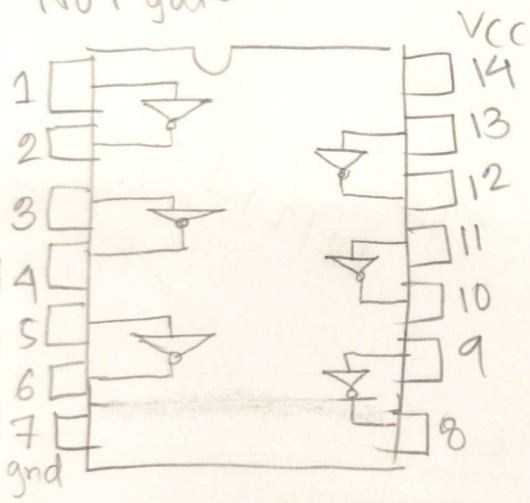
Date:

OR Gate:



input		O/P
A	B	
0	0	0
1	0	1
0	1	1
1	1	1

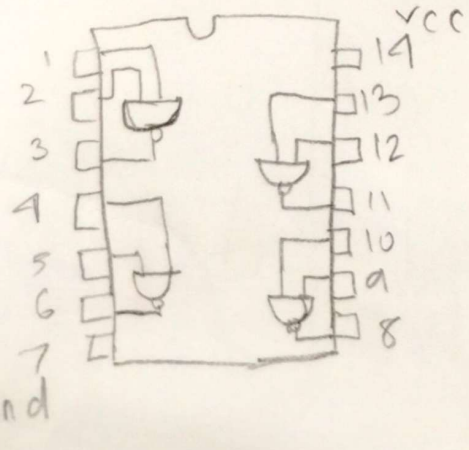
NOT gate



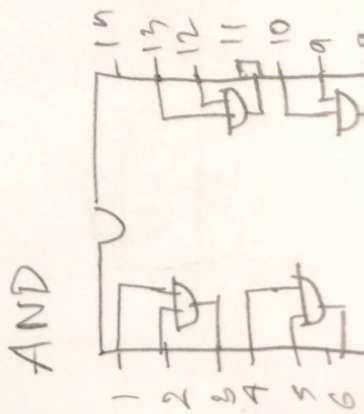
input	O/P
~~A~~ ~~GA~~	
0	1
1	0

out	0 0 0 —
B	0 — 0 —
A	0 0 — —

NAND



input		
A	B	output
0	0	1
1	0	1
0	1	1
0	0	0



Rapid Action Tablet
RABLET 20
Rabeprazole

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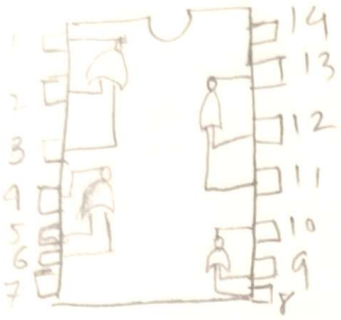
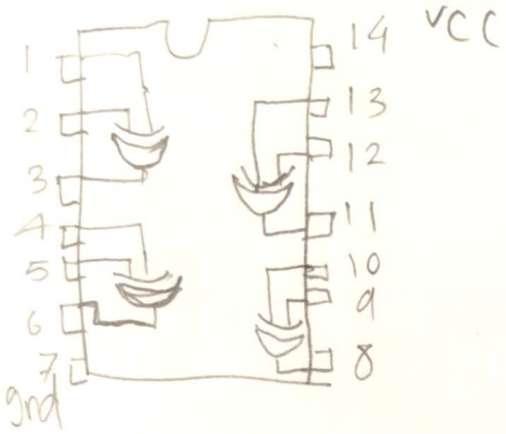
NOR



A	B	output
0	0	1
0	1	0
1	0	0
1	1	0

XOR



A	B	out
0	0	0
0	1	1
1	0	1
1	1	0

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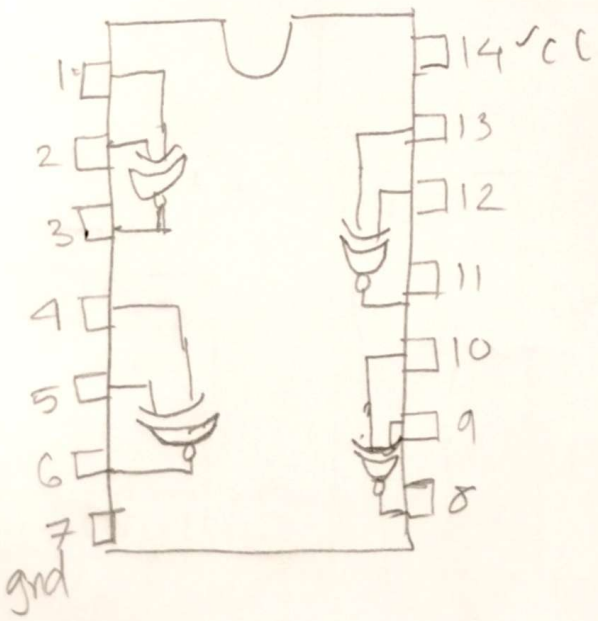
XNOR



A	B	out
0	0	1
0	1	0
1	0	0
1	1	1

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